

3 (a) (i) Initial kinetic energy of block = $\frac{1}{2}mv^2 = \frac{1}{2}(0.40)(0.30)^2 = 1.8 \times 10^{-2} \text{ J}$ **B1**

(ii) (change in) kinetic energy = work done on spring / (change in) elastic potential energy C1

$$1.8 \times 10^{-2} = \frac{1}{2} F_{MAX} (0.080)$$

$$F_{MAX} = 0.45(\text{N}) \quad \text{A1}$$

(iii) $a = F_{MAX} / m = 0.45 / 0.40$ A1
 $= 1.1 \text{ (m s}^{-2}\text{)}$

(iv) 1. constant velocity / resultant force is zero, so in equilibrium B1
2. decelerating / resultant force is not zero, so not in equilibrium B1

(b) curved line from the origin ($x_0^2 \propto m$ or $x_0 \propto \sqrt{m}$) M1
with decreasing gradient A1

4 (a) $f = 1/T = 1/0.05 = 20 \text{ Hz}$

M1